AI Project Proposal

Automated Vehicle Damage Classification for Motor Insurance Claims

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1. Introduction

This proposal presents an AI project intended for use within the motor insurance domain. It addresses the initial assessment of vehicle damage during the insurance claims process. The following sections answer the What, Why, Who, When, and How questions that frame the scope and goal of this project, before proceeding to the Domain Understanding and Model Requirements chapters.

1.1 What

This project develops an AI image-classification model that analyzes photographs of damaged vehicles, submitted as part of a motor insurance claim, and predicts the category of visible damage present in the photo (for example: scratch, dent, broken glass, bumper/body panel damage, or possible structural damage). The system takes a single image as input and returns a predicted damage category together with a confidence score.

Note on scope: the initial idea discussed with the supervisor included predicting the estimated repair cost in addition to the damage category. This is not part of the current proposal's scope: the selected dataset (Section 3.2) contains no cost information, and cost prediction would require a separate, additional data source. Repair cost estimation will only be pursued as a future extension if the damage-classification baseline proves successful and time permits; it is not a committed deliverable of this project.

1.2 Why

When a policyholder is involved in a car accident, they typically submit photographs of the damage as part of their claim. A claims handler then manually reviews these photographs to understand what type of damage occurred, which is repetitive, time-consuming, and can vary between reviewers. An AI model that performs an automatic first-pass classification of the damage shown in a photo could give claims handlers an immediate, consistent starting point before they carry out their own detailed assessment. This is a realistic and modest goal: the project does not attempt to automate the full claims process or replace human judgement, but to support the earliest step of it.

1.3 Who

The primary stakeholders are:

- Claims handlers at motor insurance companies, who would use the tool's output as a preliminary reference during claim triage.
- Insurance companies more broadly, who benefit from faster and more consistent initial claim categorization.
- Policyholders/customers, who could experience faster processing on straightforward claims.

Where possible, a domain expert (e.g. someone with experience in claims handling, vehicle assessment, or auto repair) will be involved later in the project, during Iteration 3 (Deployment & Feedback, see Section 1.4), to evaluate the tool from a real-world perspective.

1.4 When

The project will follow the AMD Project Methodology and will consist of Iteration Zero followed by three main iterations. Each iteration focuses on a different part of developing the AI solution.

Iteration Zero — Domain Understanding & Model Requirements

The goal of Iteration Zero is to create a quick first version of the solution and determine whether the project idea is feasible. The motor insurance claims domain will be explored to understand the problem, the stakeholders, and the potential value of automatically classifying visible vehicle damage.

The business objective will be translated into a machine learning problem. The target variable will be the visible damage category of a vehicle, making this a supervised multiclass classification problem. The existing vehicle-damage dataset will be used without adding additional datasets.

Iteration 1 — Data Design

Iteration 1 will focus on understanding and preparing the data. The source, quality, licensing, privacy, and suitability of the selected vehicle-damage dataset will be investigated.

Exploratory Data Analysis (EDA) will be performed to better understand the dataset. This includes looking at the number of images in each damage category, class balance, image quality, image dimensions, and other characteristics that could affect the model.

Problems found in the data, such as incorrect labels, duplicates, unusable images, or inconsistent annotations, will be investigated and cleaned where necessary. The images and labels will then be preprocessed into a suitable format for model development.

Iteration 2 — Modelling & Evaluation

Iteration 2 will focus on developing and evaluating the machine learning solution. The k-NN model from Iteration Zero will act as the initial baseline. More suitable approaches for image classification can then be explored, such as a Convolutional Neural Network (CNN) using transfer learning.

Different modelling choices can be tested and compared using appropriate classification metrics. The results will be analyzed to understand where the model performs well and where it makes mistakes. Improvements can then be made based on these findings.

The model will also be evaluated from the perspective of the original business objective: whether automatically classifying visible vehicle damage could provide useful support during the initial review of motor insurance claims.

Iteration 3 — Deployment & Feedback

The final iteration will focus on turning the selected model into a simple working prototype. Instead of interacting with the model through a development notebook, a user should be able to upload or select an image of a damaged vehicle and receive the predicted damage category.

The prototype will be demonstrated to one or more relevant stakeholders. They will be asked to test the solution and provide feedback about the predictions, usability, and potential usefulness within the vehicle damage assessment process.

The stakeholder feedback and final model results will be used to evaluate the overall project and identify possible improvements or opportunities for future development.

1.5 How

The end product is a prototype consisting of a trained image-classification model behind a simple interface. A user, such as a claims handler, uploads or selects a photograph of a damaged vehicle; the model processes the image and returns a predicted damage category together with a confidence score.

The intended final model is a Convolutional Neural Network (CNN) using transfer learning on a pretrained image model, as detailed in Section 3.1. Earlier iterations will first establish a simple baseline to confirm feasibility before this more advanced approach is developed.

2. Domain Understanding

This project sits within the domain of motor insurance claims processing, specifically the initial assessment of vehicle damage following an accident.

2.1 Research Question

To what extent can a machine-learning image-classification model accurately classify visible vehicle damage from photographs, to support the initial triage of motor insurance claims?

This question is relevant because it directly targets a real, repetitive task within the claims process (initial photo review) and produces a result that can be objectively evaluated against labeled test data — making it possible to actually answer the question with the model's measured performance.

2.2 Domain Research Findings

Research into how insurers assess vehicle damage shows that damage is generally grouped into broader categories used to decide whether a vehicle is repairable or a total loss:

- Category A / B (total loss): the vehicle is beyond economical repair and is scrapped or partially salvaged.
- Category S (structural): damage to the chassis, frame, or safety-critical components, requiring professional repair.
- Category N (non-structural / cosmetic): dents, scratches, broken glass, and other bolt-on or cosmetic damage.

A key finding from this research is that these write-off categories (A/B/S/N) are not decided from a photograph alone — they depend on additional information such as estimated repair cost, the vehicle's pre-accident market value, and physical structural inspection. For example, insurers commonly treat a vehicle as an economic total loss once repair costs reach roughly 60–70% of its market value, which is not information a single image can provide.

2.3 Scope and Added Value

Because a full write-off classification is not feasible from image data alone, this project scopes itself to a task that a photograph can meaningfully answer: classifying the type of visible damage shown in the image (e.g. scratch, dent, broken glass, bumper/body damage, possible structural damage). This keeps the target variable realistic and measurable, while still providing genuine value to the stakeholder — a fast, consistent first read of what a submitted photo shows, before a human reviewer takes over for any decision involving cost or write-off status.

3. Model Requirements

3.1 Analytic Approach

The business objective of this project is to support the initial review of vehicle damage photographs during the motor insurance claims process. This objective is translated into a machine learning problem where a model predicts the type of visible vehicle damage shown in an image.

Target variable: The category of visible damage shown in a vehicle photograph.

Nature of the problem: Supervised multiclass classification. The dataset contains labeled examples of vehicle damage, and the goal is to predict the correct damage category for new, unseen examples.

The damage categories available in the selected dataset are Dent, Scratch, Crack, Glass Shatter, Lamp Broken, and Tire Flat.

3.1.1 Choice of Algorithm: CNN via Transfer Learning

Training a CNN from scratch typically requires tens or hundreds of thousands of images to learn reliable visual features. The selected dataset (4,000 images, see Section 3.2), while larger than many small-scale alternatives, is still modest compared to what training a network from scratch would require, so this approach would still likely lead to weaker performance and overfitting than starting from a pretrained model.

To address this, the project will use transfer learning: starting from a CNN that has already been pretrained on a large, general-purpose image dataset (such as ImageNet), and adapting it to the specific task of vehicle damage classification. The pretrained model has already learned general visual features (edges, textures, shapes, object parts) from millions of images; only the final layer(s) of the network need to be retrained to recognize the project's 6 damage categories. This makes transfer learning realistic and appropriate for a dataset of this size.

Candidate pretrained models to evaluate:

- MobileNetV2 — a lightweight, fast architecture, well suited to a prototype that needs to run inference quickly, e.g. in a browser-based demo.
- ResNet (e.g. ResNet18/ResNet50) — a widely used, well-documented baseline in image classification with strong general performance.
- EfficientNet — a modern architecture that tends to achieve high accuracy relative to its size and computational cost.

The project will begin with MobileNetV2 or ResNet18 as a baseline model, given their strong documentation and manageable computational requirements, and compare results against alternative pretrained models if time allows during Iteration 4 (Model refinement).

Traditional machine learning algorithms (e.g. decision trees, logistic regression, SVM) were considered but rejected for this task, because they require manually engineered features and are not well suited to learning directly from raw image pixels. CNNs remove this limitation by learning visual features automatically, which matches the nature of this classification problem.

Good indicators: in image classification, relevant "features" are visual patterns learned automatically by the model — edges, shapes, textures, and color patterns associated with each damage type — rather than hand-selected columns as in tabular data. Exploratory analysis will focus on inspecting the image dataset for consistency (resolution, lighting, angle, class balance) rather than calculating correlation coefficients.

Damage categories, based on the confirmed dataset (see Section 3.2):

- Dent
- Scratch
- Crack
- Glass shatter
- Lamp broken
- Tire flat

3.2 Data Requirements

This section defines what data is needed to train and evaluate the model.

Data type: unstructured data — photographs of damaged vehicles, each associated with a damage-category label.

Labeling: since this is supervised learning, labeled data is required. Each image must have a ground-truth damage category assigned, ideally by the original dataset's annotators.

Selected dataset: CarDD (Car Damage Detection), a dataset published by researchers at the Chinese Academy of Sciences and the University of Science and Technology of China (Wang, Li & Wu, 2023, IEEE Transactions on Intelligent Transportation Systems). It contains 4,000 high-resolution car damage images with over 9,000 annotated damage instances, purpose-built for vision-based car damage detection and segmentation.

Classes: the dataset is annotated with 6 damage categories: Dent, Scratch, Crack, Glass Shatter, Lamp Broken, and Tire Flat. These categories were selected by the dataset's authors based on frequency of occurrence in real insurance claims, making them a realistic fit for this project's insurance-claim use case.

Data source: the dataset is published by its original academic authors, with the paper, license, and download instructions available at the official project page (cardd-ustc.github.io); the same image files are also mirrored on Kaggle for direct download, avoiding delays while the official licensing request is processed.

Legality and ethics: access to the original dataset requires submitting a licensing form to the authors, permitted for non-commercial academic and educational use, which this student project satisfies; the Kaggle mirror will be used for practical access to the image files. As a published, peer-reviewed dataset, it can be properly cited in the project report. No personally identifiable information (e.g. license plates, faces) is a primary subject of the images, but this will still be checked during preprocessing.

Volume and split: with 4,000 images and over 9,000 annotated instances across 6 classes, this dataset is substantially larger than typical small-scale alternatives, giving more headroom for training a CNN via transfer learning. Given the 3-month project timeline, this additional volume is manageable and will support a more reliable train/validation/test split (e.g. 70/15/15) with enough examples per class. Data augmentation may still be used to further improve robustness.

Diversity: the dataset's images were purpose-collected for car damage detection and cover a range of damage severities and vehicle viewpoints; during EDA the dataset will be reviewed for class balance across the 6 categories and for variety in angle, lighting, and vehicle type.

Because CarDD provides instance-level annotations (multiple damage instances can be annotated per image) rather than a single label per image, a data-preparation step is still required before training a single-label classifier: each image will be assigned a dominant-class label (the most prominent or largest annotated damage instance). This conversion approach should be piloted early in the Data phase (Iteration 1).